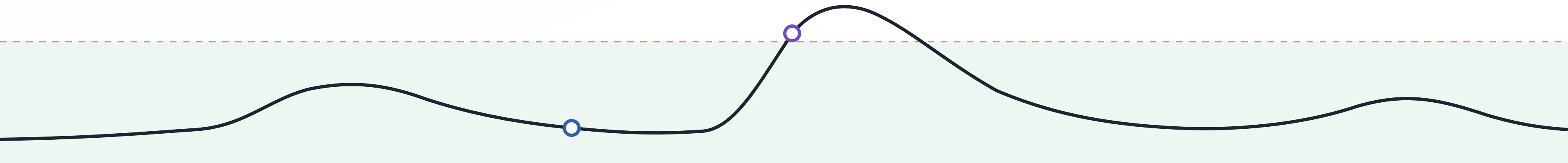


Glycemic Twin

A clinician-facing Digital Twin that estimates, at each meal, the risk that glucose will exceed 180 mg/dL within two hours, for prediabetes and type 2 diabetes.

Built on real CGMacros v1.0.0 data (PhysioNet) · Team leader: **Mitansh Mathur**



TEAM

IdeaForge

Team name

IdeaForge

Team leader

Mitansh Mathur

College

IIIT Allahabad

What we built, end to end

Data audit → leakage-controlled feature engineering → evaluated ML models → a deterministic Digital Twin engine → a secure FastAPI + PostgreSQL backend → an interaction-first Next.js clinician interface → Docker, Caddy and CI. All on real, open data.

THE PROBLEM

Post-meal glucose spikes are personal, and today's tools only show them afterwards

- In prediabetes and type 2 diabetes, glucose often rises **above 180 mg/dL** after meals, the upper end of the usual 70–180 target range.
- The response depends on the **person**, not only the meal: pre-meal glucose and trend, activity, metabolic baseline, past responses.
- CGMs record **what already happened**. A reviewing clinician cannot see which meals were likely to spike, or why, before the outcome.

The question a clinician asks

“Given everything known about this person **up to now**, how likely is this meal to take them above 180 within two hours, **why**, and how would a **different meal** look to the model?”

Decision support for reviewing a patient's meals and glucose

Who

Clinicians: diabetes educators, endocrinologists and primary care, reviewing CGM, meal and wearable data before or during follow-up.

For whom

People with **prediabetes** or **type 2 diabetes** (groups derived from baseline HbA1c). Healthy participants are evaluated separately.

What it gives

Per-meal risk with uncertainty, its drivers, the person's learned response, prediction vs. reality, bounded what-if, and replay of how the Twin evolved.

Deliberately out of scope

No medication, insulin, diagnosis or treatment functionality. No automated actions. Not a medical device, not clinically validated, and not live monitoring.

Why current approaches are insufficient

APPROACH	WHAT IT DOES	WHAT IS MISSING
CGM report	Shows curves and time in range afterwards	No estimate before the outcome, no reasons, no scenarios
Population rules	“Fewer carbs”, fixed carb limits	Ignores how this person responds
A single ML score	One number per meal	No history, personalization, uncertainty or traceability; often evaluated with leakage
Glycemic Twin	A per-person state at any moment, which predicts and explains	Personal, time-honest, explainable, traceable, queryable with bounded what-if

A Digital Twin is a living per-person state, not a single score

```
build_state(patient, as_of) → TwinState
```

Built only from what was observable **up to** **as_of**. Nothing later is visible, and a meal's outcome counts only once its 2-hour window has closed.

Mirrors

clinical baseline, recent glucose
and heart rate, meals, personal
response

Updates

rebuilt at any moment; evidence
accumulates as time moves
forward

Predicts & explains

calibrated risk, drivers,
uncertainty, full provenance

Can be queried

bounded what-if; diff_states
explains change

COLD_START



WARMING



PERSONALIZED

PERSONALIZED: own-evidence weight ≥ 0.5 and ≥ 3 closed meals. Not a 3D avatar: it mirrors glycemic state and behaviour.

CGMacros v1.0.0: real free-living data, about 10 days per participant

44

participants with time series (14 healthy, 16 prediabetes, 14 T2D)

1,663

logged meals with macronutrients

123k

native Dexcom readings (5-min)

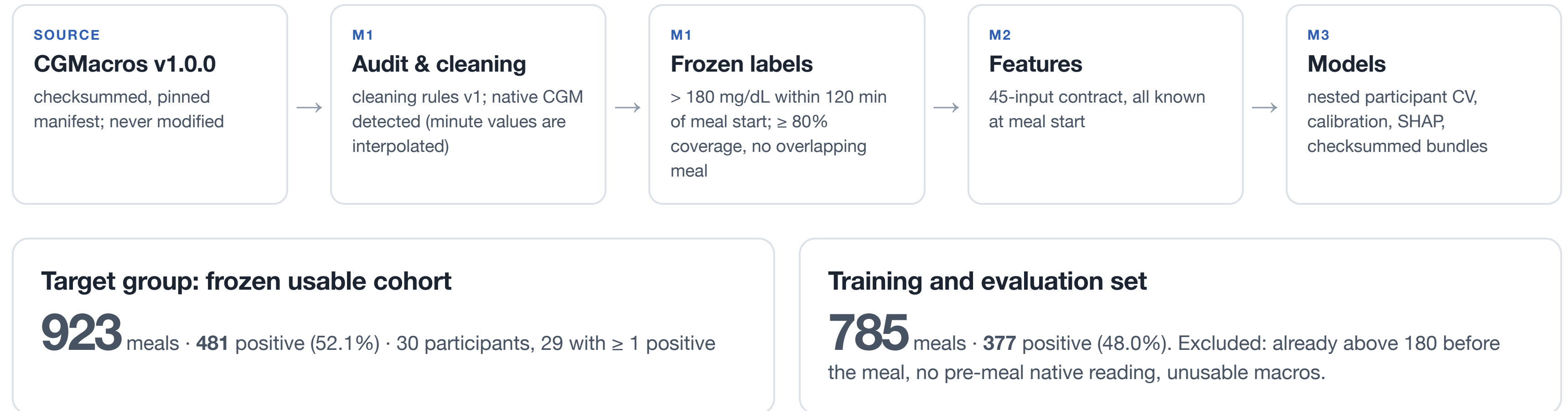
Attribution

Gutierrez-Osuna, R., Kerr, D., Mortazavi, B., & Das, A. (2025). *CGMacros: a scientific dataset for personalized nutrition and diet monitoring* (v1.0.0). PhysioNet. doi:10.13026/3z8q-x658

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- **Dexcom G6 CGM, Fitbit** heart rate and activity, **meal macros, baseline labs** (HbA1c, fasting glucose, lipids...).
- Groups are **derived from HbA1c**; the dataset has no diagnosis or medication fields, and none are invented.

Every stage leaves an auditable, versioned report



45 inputs, nested participant-level cross-validation

GROUP	#	INPUTS (ALL KNOWN AT MEAL START)
Pre-meal glucose	11	last native reading, slopes, variability, 3-h mean/min/max, time in range, overnight baseline
Meal	11	carbs, protein, fat, fiber, calories, meal type, time of day
Context	3	recent carbs, minutes since the last meal
Wearable	8	heart rate vs resting, METs, active minutes
Clinical	9	HbA1c, fasting glucose and insulin, HOMA-IR, BMI, lipids
Personal	3	shrinkage estimates from the person's earlier closed meals

- **XGBoost** and **logistic regression**, both calibrated; threshold chosen on inner folds.
- **5 pinned, participant-disjoint folds**; tuning, calibration, threshold and personal prior never see the test participant.
- 4 baselines and modality ablations; participant-bootstrap 95% CIs and paired comparisons.

The model never sees the answer, and automated checks prove it

L2 • no future data

0 meals read data after their start time.

L3 • recomputation

Features rebuilt from history cut at meal start are identical.

L4 • disjoint folds

No participant appears in both training and test.

L5 • clean windows

No meal inside another usable meal's 2-hour window.

L6 • sanity

No single feature is suspiciously predictive.

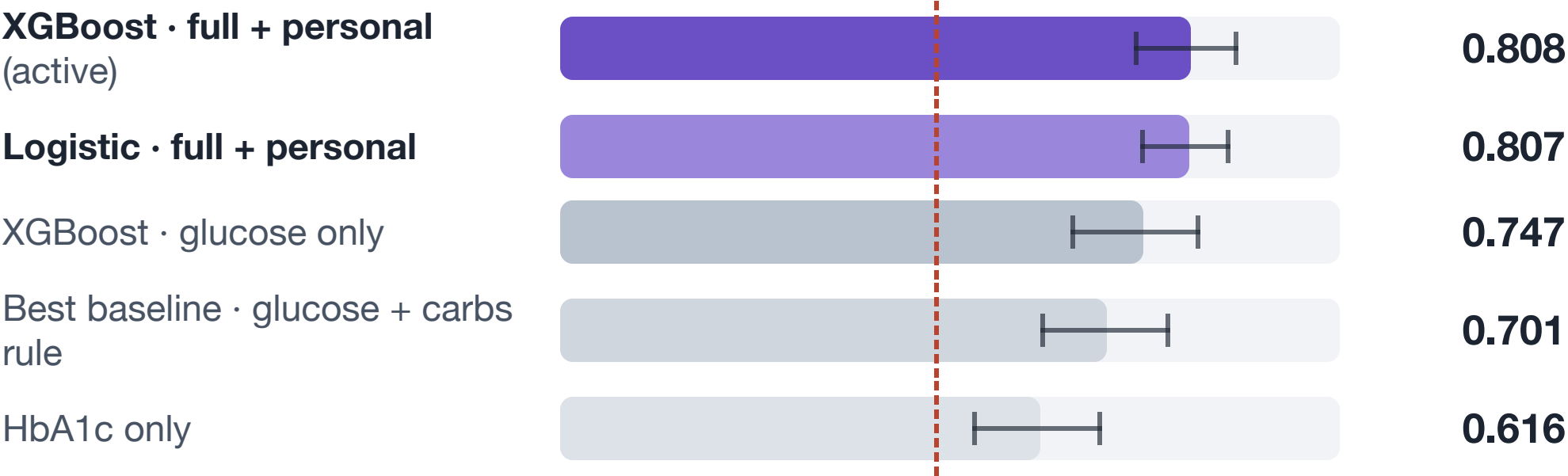
Personal prior

Fitted per fold, on training participants only.

“**Amount Consumed**” is logged after the meal, so it is **never** a feature. Labels and features are kept on separate bases (ADR-013). In the Twin, tests append different future data and check that the state hash **does not change**.

Clear gain over baselines; XGBoost and logistic regression are equivalent

PR-AUC with 95% CI · target group, 785 meals · dashed line = prevalence 0.480 (random model)



Active model: AUROC 0.804 [0.759, 0.845] · Brier 0.179 · calibration slope 1.00 · at the inner-fold threshold: sensitivity 0.806, specificity 0.630

vs best baseline

+0.107 PR-AUC [0.059, 0.158] · +0.064 AUROC (paired bootstrap)

XGBoost vs logistic

+0.001 [−0.034, 0.038]: no meaningful difference. Not claimed superior.

Fusion · personalization

Multimodal vs glucose-only **+0.048** [0.010, 0.091]. Personal prior +0.013 [−0.017, 0.041]: not proven.

One pure-Python core shared by training and serving

1 · TIME BOUNDARY

Cut every observation at as_of

and close only meal windows that finished before it



2 · SAME CODE AS TRAINING

42 base + 3 personal inputs

prior and calibrator come from the model bundle; contract and config hashes checked (no train/serve skew)



3 · PREDICT

Calibrated risk + threshold + uncertainty

model bundle SHA-256 verified on load



4 · TWINSTATE

Immutable, content-hashed, versioned

lifecycle, drivers, personal response, freshness, provenance (input hashes, configs, model identity)

simulate()

bounded, non-causal what-if on the current meal

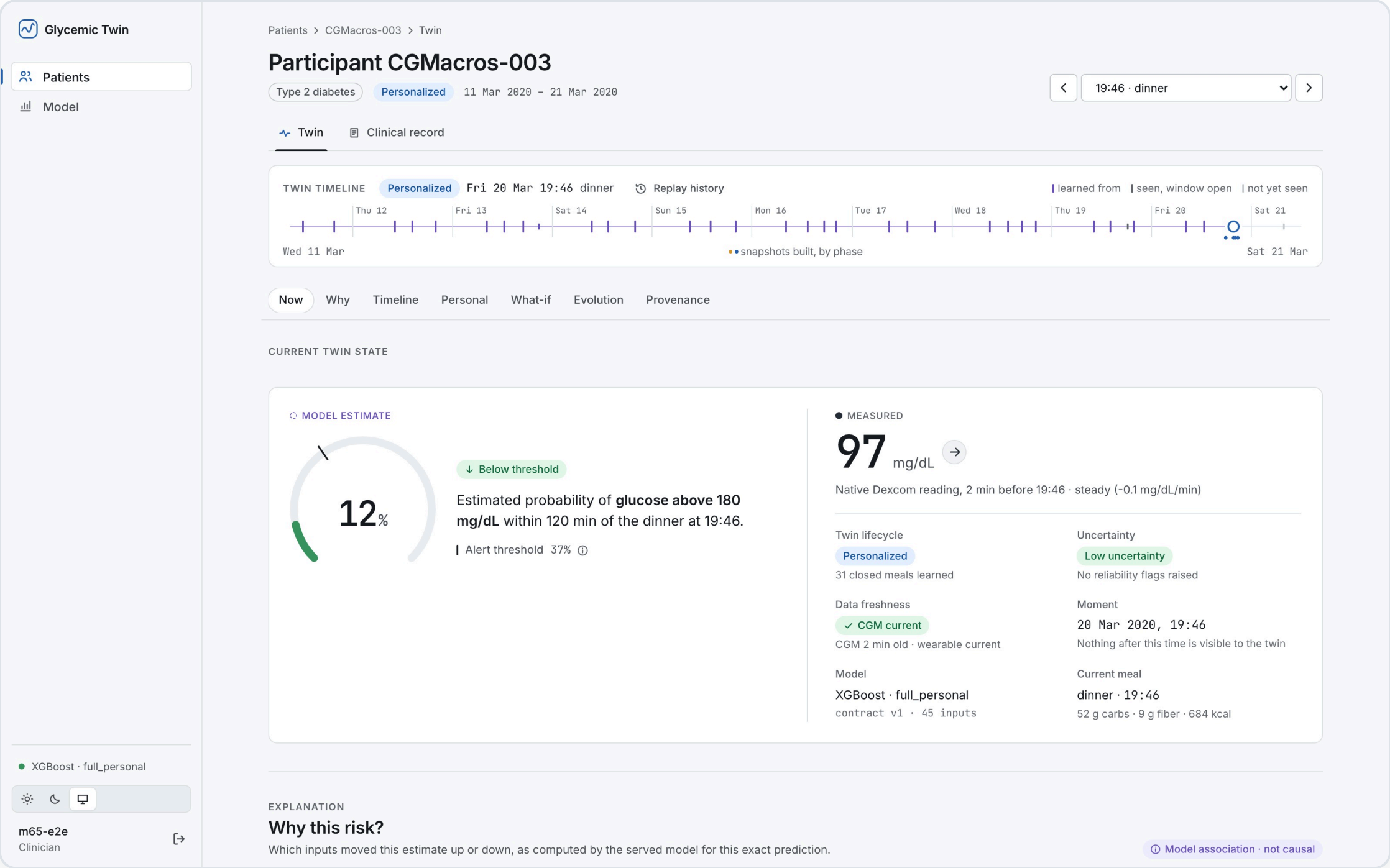
diff_states()

what changed between two moments, and why

Deterministic

same data + same moment → same state and hash. A test rebuilds all 45 inputs through the Twin and matches the training data exactly.

Calm at rest, alive on interaction



- **Risk dial** counts to the calibrated estimate against the alert threshold.
- **Measured glucose:** the chevron opens the 90-minute CGM context in place.
- **Twin timeline:** learned from / window open / not yet seen.
- Lifecycle, uncertainty, data freshness, model and moment are always visible.

Every estimate shows what it rests on, and can be checked against what happened

EXPLANATION

Why this risk?

Which inputs moved this estimate up or down, as computed by the served model for this exact prediction.

lowers the estimate

raises the estimate

bar length = share of the largest contribution

Glucose at meal start

GLUCOSE

97 mg/dL

Last native Dexcom reading within 15 min before the meal. Associated with a lower estimate for this meal.

Computed from the native glucose readings in the 15 min before the meal.

Show on timeline →

Protein

THIS MEAL

28 g

Logged protein of this meal. Associated with a lower estimate for this meal.

Carbohydrate

THIS MEAL

52 g

Logged carbohydrate of this meal. Associated with a lower estimate for this meal.

Taken from this meal's log.

Personal rise offset

PERSONAL RESPONSE

12.1 mg/dL

How much more (or less) this person's glucose rose than the population model expects. Associated with a higher estimate for this meal.

Lowest glucose, 3 h

GLUCOSE

97 mg/dL

Minimum reading in the last 3 hours. Associated with a lower estimate for this meal.

Overnight baseline

GLUCOSE

110 mg/dL

Median glucose 02:00–06:00 on completed nights. Associated with a lower estimate for this meal.

The other 39 inputs together add +0.91 and -1.30.

Show all 45 inputs

Model baseline -0.45 → + inputs -1.94 → raw score -2.39 log-odds → calibrated 12%

Exact TreeSHAP on the served model. Contributions to the model's raw score (log-odds, before calibration). They show what the model relied on for this prediction; they are associations, not caus

OBSERVED

Glucose and meals

Native Dexcom readings around the meal. The twin sees only what was recorded before its moment; reveal the rest to compare prediction with reality.

Model: 12% chance > 180

180 mg/dL · outcome threshold

twin's view ends

16:00

17:00

18:00

19:00

20:00

21:00

22:00

Dinner at 19:46

current meal

Carbs 52 g

Fiber 9 g

Protein 28 g

Fat 43 g

Energy 684 kcal

Shaded: the 2 hours after it (readings after the twin's moment stay hidden).

Measured (native Dexcom, 5-min)

Measured later, hidden from the twin

Prediction window, 120 min

70–180 mg/dL

Meal

What actually happened after 19:46?

The twin predicted 12% using only data up to 19:46. Reveal the next 120 minutes of recorded glucose to compare, or drag through them.

Reveal what happened

- Exact TreeSHAP on the served model: raised / lowered, association not cause.
- The chart shows only what the Twin saw; “Reveal what happened” compares prediction with the recorded outcome.
- Provenance: model version, contract, data and config hashes.

Change the meal, and see the model's estimate move from its baseline

SIMULATION

What if this meal were different?

Change the logged macros of the current meal and see how the model's estimate would move. Bounded to the range the model was trained on.

Model-estimated scenario

Non-causal

Not medical advice

Try

-20 g carbs

+10 g fiber

+30 active min

Carbohydrate

52 → 32 g

-20 g

Fiber

9 → 9 g

±0 g

Protein

28 → 28 g

±0 g

Fat

43 → 43 g

±0 g

Active minutes before the meal

63 → 63 min

±0 min

Calories follow macro changes automatically. Post-meal activity cannot be simulated: the model has no post-meal inputs.

Reset

AS LOGGED

12%

52 g carbs

SCENARIO

11%

32 g carbs

-0.7 pp

As logged

Scenario

12%

11%

Carbohydrate 52 → 32 g

Stays below the alert threshold, further to it

25.8 pp below the 37% threshold (was 25.1 pp below)

Within training support

Low uncertainty

Held fixed: every other input: glucose history, wearable, clinical values, meal time and type, and the personal features (they depend only on closed past meals).

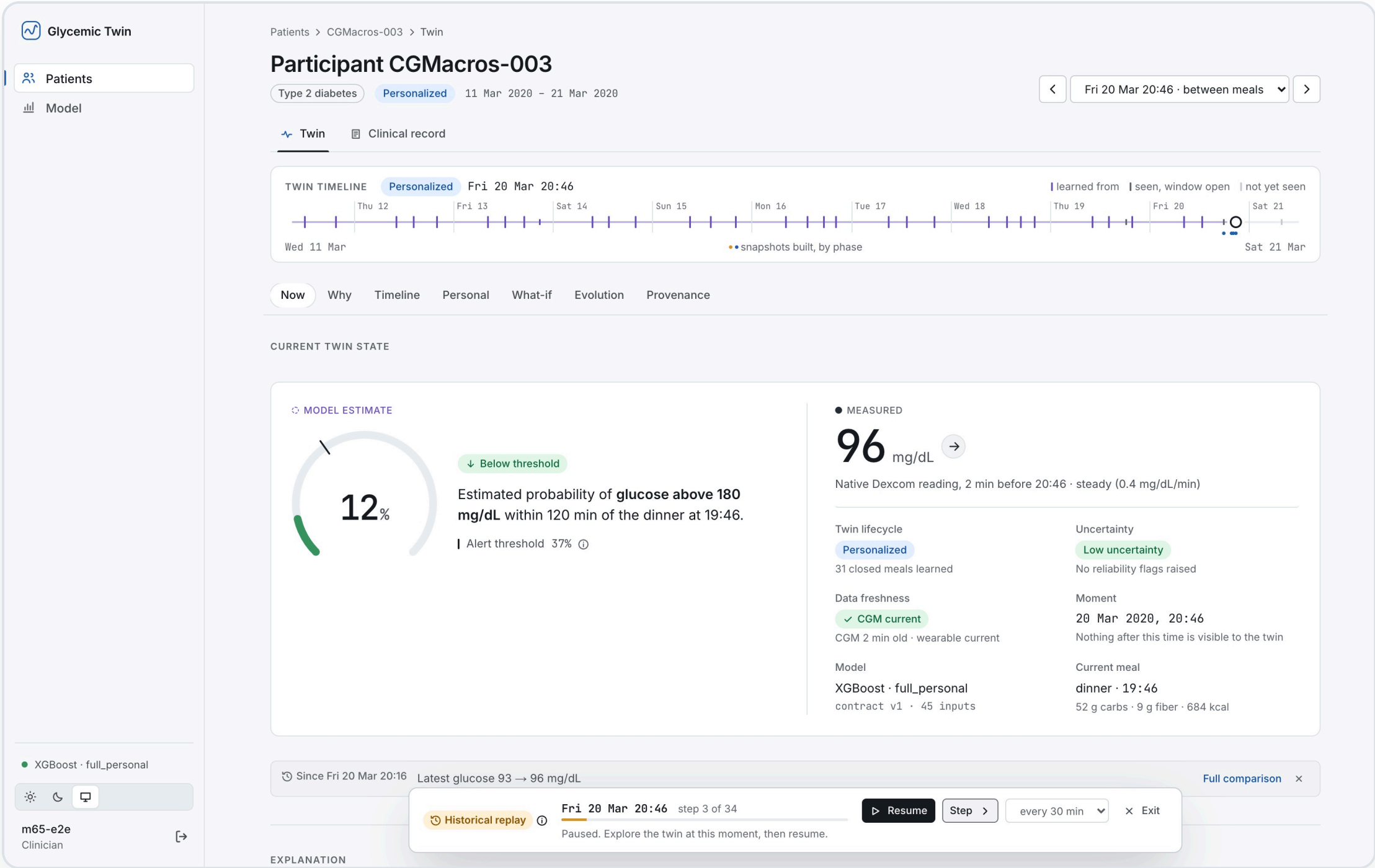
- Levers: **carbs, fiber, protein, fat** of the current meal, and **active minutes before it**.
- Baseline stays visible → estimate transitions → **delta**, threshold crossing, uncertainty.
- Bounded**: inside the training 1st–99th percentile, or “out of support” with no number.

Model-estimated

Non-causal

No medication / insulin levers

Watch the Twin evolve through a patient's recorded days



- **Play, pause, step;** at each step the server rebuilds the Twin at the new as_of.
- Only information up to that moment is used; closed meals become personal evidence.
- The **time rail** moves the Twin through history the same way.
- **Historical replay, not live monitoring. Not a held-out evaluation** (the serving model was refit on all participants).

Clinical-grade habits in a research prototype

Authentication

Argon2id passwords, lockout after 5 failures, no user enumeration. Google sign-in only for accounts an admin provisioned.

Sessions

Server-side, hashed 256-bit tokens; HttpOnly, SameSite, Secure cookies; CSRF token on every change.

Roles and audit

Clinician / admin; sign-ins, denials, model activation and user creation are audited, never with secrets.

Production guards

Refuses to start without secure cookies and trusted hosts; documentation endpoints off; generic errors.

Logs

JSON with request IDs and secret redaction; no bodies, cookies or query strings.

Data

De-identified research data only. No raw data, patient-level tables, model bundles or secrets in git.

A small, defensible stack

LAYER	TECHNOLOGY
Data and ML	Python 3.11, pandas, PyArrow, scikit-learn, XGBoost, SHAP; uv lock file
Twin core	twin_core: pure Python, no web or database code
Backend	FastAPI, Pydantic v2, SQLAlchemy 2, Alembic, PostgreSQL 16
Frontend	Next.js 15, React 19, TypeScript, TanStack Query, custom SVG charts
Delivery	Docker (non-root), Compose, Caddy (HTTPS, one origin), GitHub Actions

Verified quality

370 backend tests (incl. PostgreSQL migrations) · **105** frontend unit tests · **51** Playwright E2E with axe accessibility, reduced motion and mobile

CI on every pull request

lint, types, tests (fails if any test is skipped), production images built for linux/amd64

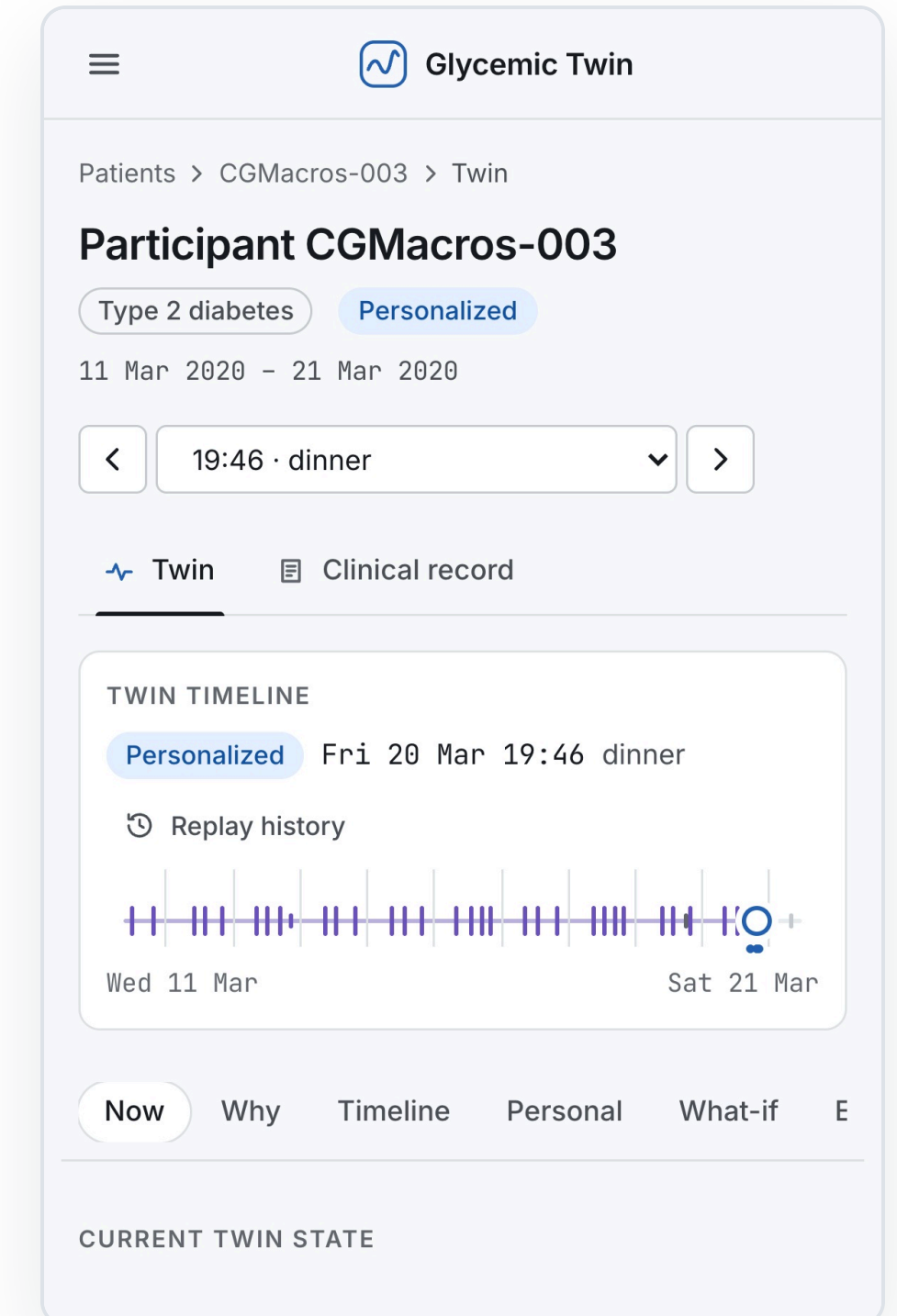
No LLMs

No chatbot and no external AI APIs.

OUTCOME

A complete, reproducible prototype from open data to a clinician interface

- Real data, audited and cleaned with versioned rules; **leakage checks all passed**.
- A model that beats simple baselines on **unseen participants** (PR-AUC 0.808 vs 0.701), well calibrated.
- A Digital Twin with an honest **time boundary**, lifecycle, provenance, what-if and replay.
- A secure backend and an accessible, responsive, interaction-first interface.
- One-command Docker stack and green CI; MIT-licensed code.



LIMITATIONS

What this prototype cannot claim

Small cohort

About 30 target-group participants over ~10 days: wide confidence intervals.

One evaluation regime

New participants only; temporal validation is future work.

Association, not causation

What-if and SHAP show what the model relies on, not what would happen to a person.

Label basis

Labels use the frozen 1-minute series; the effect of interpolation is unmeasured (SA-1).

Narrow levers

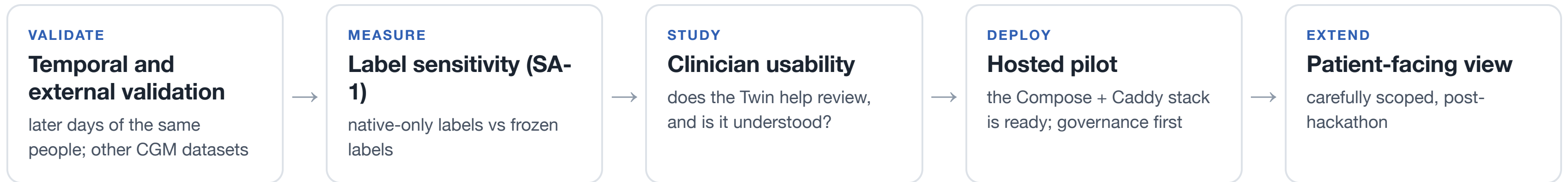
Current-meal macros and pre-meal activity only; no post-meal activity, medication or insulin.

Not clinical

Not a medical device, not clinically validated, not deployed publicly, not live monitoring.

FUTURE SCOPE

Next steps, in order of evidence first



Let's see the Twin.

Sign in → patient list → Patient Twin → glucose and meals → prediction vs reality → what-if → time rail and historical replay → provenance.

Code

github.com/mitanshm18/glycemic-twin

License

Code MIT · Data CC BY-NC-SA 4.0

Team

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